

◆ Architecture Exercise

AWS Solutions Architect Associate (SAA-C03)

Topic:

EC2 Instance Types • High Availability • Fault Tolerance • Scaling Concepts

◆ Real SAA Exam Scenario

A company hosts a highly popular e-commerce website on AWS.

The company experiences unpredictable traffic spikes during special promotions and Black Friday sales.

Management requires the following:

- ✓ High Availability
- ✓ Fault Tolerance
- ✓ Automatic Scaling
- ✓ Cost Optimization
- ✓ Secure Architecture

As the AWS Solutions Architect, design a solution that meets these requirements.

◆ Step 1 — Understand the Requirements

Before selecting AWS services, identify what the business needs.

Requirement 1

Application must remain available if a server fails.

Requirement 2

Application must automatically handle traffic spikes.

Requirement 3

Application should avoid paying for unused resources.

Requirement 4

Database must remain available during failures.

Requirement 5

Architecture should follow AWS best practices.

◆ Step 2 — Select the Correct AWS Services

Compute Layer

Use:

- Amazon EC2

Purpose:

Run the application servers.

Load Balancing Layer

Use:

- Application Load Balancer (ALB)

Purpose:

Distribute incoming traffic across multiple EC2 instances.

Scaling Layer

Use:

- Auto Scaling Group (ASG)

Purpose:

Automatically launch or terminate EC2 instances based on demand.

Database Layer

Use:

- Amazon RDS Multi-AZ

Purpose:

Provide automatic database failover.

Networking Layer

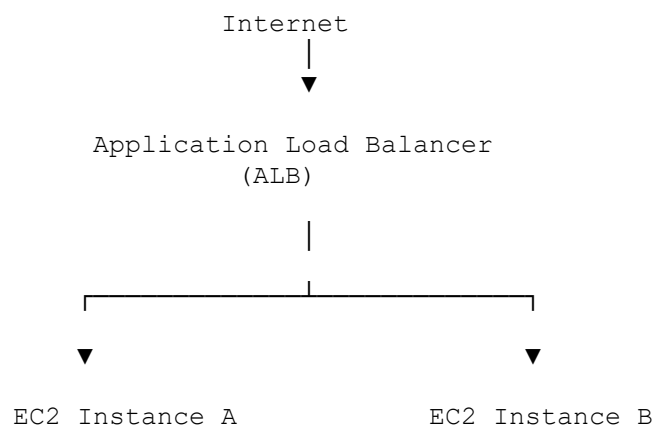
Use:

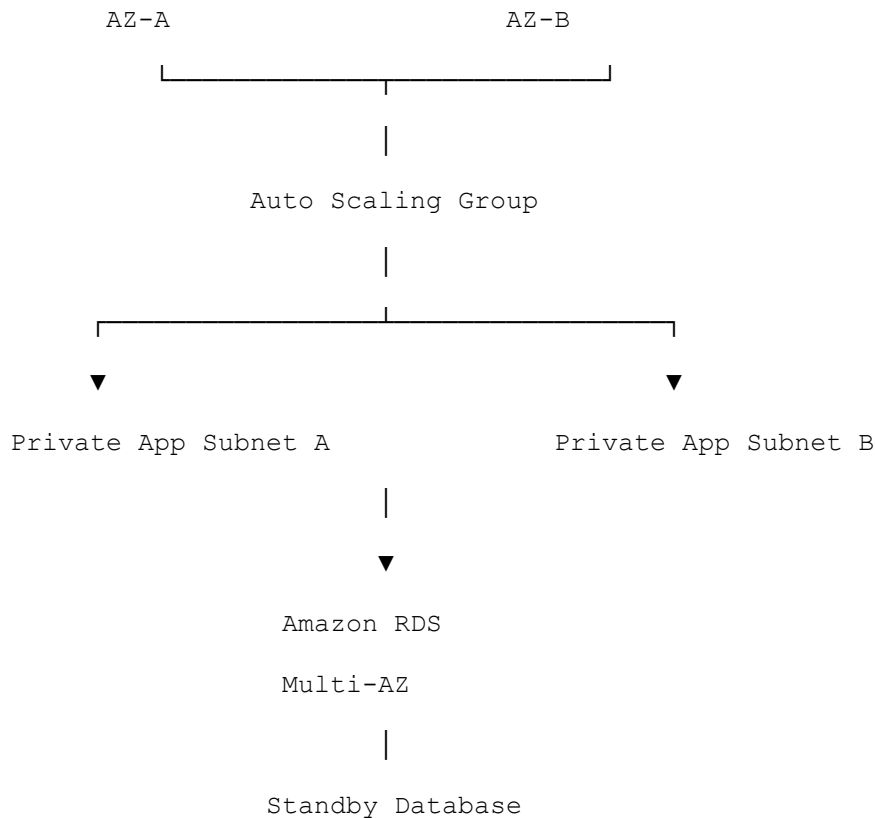
- VPC
- Public Subnets
- Private Subnets
- Internet Gateway
- NAT Gateway

Purpose:

Create a secure network architecture.

◆ Step 3 — Recommended Architecture





◆ High Availability Analysis

What is High Availability?

The ability of a system to remain available despite failures.

How Is High Availability Achieved?

Using:

- Multiple Availability Zones
 - Load Balancers
 - Auto Scaling
 - Multi-AZ Databases
-

Example

If AZ-A experiences an outage:

AZ-A ✖ Failed

AZ-B ✓ Available

The application continues running.

Users remain connected.

◆ Fault Tolerance Analysis

What is Fault Tolerance?

The ability to continue operating when components fail.

Example

If one EC2 instance crashes:

EC2-A ✗ Failed

EC2-B ✓ Healthy

The Application Load Balancer automatically routes traffic to healthy servers.

No manual intervention is required.

SAA Exam Tip

High Availability reduces downtime.

Fault Tolerance eliminates service interruption.

◆ Scaling Analysis

Vertical Scaling

Also known as:

Scale Up

Example:

t3.micro

↓

t3.large

More CPU and memory are added.

Horizontal Scaling

Also known as:

Scale Out

Example:

2 EC2 Instances

↓

10 EC2 Instances

More servers are added.

Which Scaling Method Does AWS Prefer?

AWS generally prefers:

✓ Horizontal Scaling

Because it improves:

- Availability
 - Fault Tolerance
 - Elasticity
-

◆ Instance Type Selection

General Purpose

Examples:

- T3
- T4g

- M5

Best For:

- Web Applications
 - Development Environments
-

Compute Optimized

Examples:

- C5
- C6i

Best For:

- Scientific Computing
 - Machine Learning
 - CPU-intensive Workloads
-

Memory Optimized

Examples:

- R5
- R6i

Best For:

- Databases
 - SAP HANA
 - In-Memory Applications
-

Storage Optimized

Examples:

- I3
- D2

Best For:

- Big Data

- Data Warehouses
-

◆ SAA Exam Questions

Question 1

A company wants to distribute incoming traffic across multiple EC2 instances.

Which AWS service should be used?

- A. Route 53
- B. CloudFront
- C. Application Load Balancer
- D. Auto Scaling

Answer

✓ C. Application Load Balancer

Question 2

A company requires automatic database failover.

Which solution should be used?

- A. Read Replica
- B. EBS Snapshot
- C. Multi-AZ RDS
- D. Security Group

Answer

✓ C. Multi-AZ RDS

Question 3

A company increases an EC2 instance from t3.micro to t3.large.

What type of scaling is this?

- A. Horizontal Scaling
- B. Vertical Scaling
- C. Auto Scaling
- D. Elastic Scaling

Answer

✓ B. Vertical Scaling

Question 4

A company needs EC2 instances automatically added when CPU utilization exceeds 70%.

Which service should be used?

- A. CloudWatch
- B. Route 53
- C. Auto Scaling Group
- D. IAM

Answer

✓ C. Auto Scaling Group

Question 5

Which architecture provides the highest availability?

- A. One EC2 instance in one Availability Zone
- B. Two EC2 instances in one Availability Zone
- C. Multiple EC2 instances across multiple Availability Zones
- D. One large EC2 instance

Answer

✓ C. Multiple EC2 instances across multiple Availability Zones

◆ SAA Exam Memory Tricks

Instance Types

Remember:

G-C-M-S

Letter	Meaning
G	General Purpose
C	Compute Optimized
M	Memory Optimized
S	Storage Optimized

Scaling

Scale Up = Bigger Server

Scale Out = More Servers

Availability

High Availability
=
Stay Online

Fault Tolerance

Fault Tolerance
=
Continue Operating After Failure

◆ Key Exam Takeaways

Before the exam, remember:

✓ General Purpose = Balanced Workloads

- ✓ Compute Optimized = CPU Intensive
 - ✓ Memory Optimized = Database Workloads
 - ✓ Storage Optimized = Big Data
 - ✓ High Availability = Multiple AZs
 - ✓ Fault Tolerance = Survive Failures
 - ✓ Vertical Scaling = Scale Up
 - ✓ Horizontal Scaling = Scale Out
 - ✓ Auto Scaling = Automatic Capacity Adjustment
 - ✓ Load Balancer = Traffic Distribution
 - ✓ Multi-AZ RDS = Database Failover
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Exam Focus: If you understand why AWS recommends **Multi-AZ + Load Balancer + Auto Scaling**, you have mastered one of the most tested architecture patterns in the AWS Solutions Architect Associate exam.